

# SCADA System

## Recurring Cost Estimation

---

Augmentation of Water Supply Scheme  
Burhar Nagar Parishad, Distt. Shahdol  
**Under AMRUT 2.0**

PLC, SCADA & Instrumentation  
Real-Time Monitoring & Control System

*Prepared by: Orbit Automation  
Date: March 2026*

## 1. Project Overview

This document provides a detailed recurring cost estimation for the SCADA (Supervisory Control and Data Acquisition) system deployed under the Augmentation of Water Supply Scheme, Burhar Nagar Parishad, District Shahdol, under AMRUT 2.0.

The SCADA system covers three primary sections:

- Intake Section — Level, Pressure (PT), Flow, Pumps
- WTP (Water Treatment Plant) — Level, Pressure (6 PT), Flow, Chlorine, pH, Turbidity, Pumps
- OHT (Overhead Tank) — Level, Flow (Inlet/Outlet)

Cost estimates are provided in three tiers — Low, Medium, and High — to give flexibility in budgeting. Costs are shown for 1 to 5 year durations.

## 2. Local Infrastructure Costs (Per Year)

These are on-site recurring costs for hardware, networking, manpower, and maintenance.

| Component                   | Low             | Medium            | High              |
|-----------------------------|-----------------|-------------------|-------------------|
| Server/Hardware Maintenance | ₹ 3,000         | ₹ 6,000           | ₹ 12,000          |
| Internet & Networking       | ₹ 6,000         | ₹ 9,000           | ₹ 15,000          |
| AMC (Annual Maintenance)    | ₹ 12,000        | ₹ 24,000          | ₹ 36,000          |
| Spare Parts & Instruments   | ₹ 5,000         | ₹ 10,000          | ₹ 18,000          |
| Manpower / Operator         | ₹ 24,000        | ₹ 48,000          | ₹ 72,000          |
| Miscellaneous               | ₹ 3,000         | ₹ 6,000           | ₹ 10,000          |
| <b>Total / Year</b>         | <b>₹ 53,000</b> | <b>₹ 1,03,000</b> | <b>₹ 1,63,000</b> |

## 3. Cloud & Software Costs (Per Year)

These cover cloud hosting, data storage, alerting, domain, and remote technical support.

| Component                   | Low             | Medium          | High            |
|-----------------------------|-----------------|-----------------|-----------------|
| Cloud Hosting (Server)      | ₹ 3,600         | ₹ 7,200         | ₹ 14,400        |
| Domain & DNS                | ₹ 1,000         | ₹ 1,500         | ₹ 2,000         |
| SMS/Email Alerts            | ₹ 1,200         | ₹ 3,600         | ₹ 6,000         |
| Data Storage & Backup       | ₹ 1,200         | ₹ 3,000         | ₹ 6,000         |
| Technical Support & Updates | ₹ 6,000         | ₹ 12,000        | ₹ 24,000        |
| SSL Certificate             | ₹ 0             | ₹ 0             | ₹ 0             |
| <b>Total / Year</b>         | <b>₹ 13,000</b> | <b>₹ 27,300</b> | <b>₹ 52,400</b> |

## 4. Total Recurring Cost Summary (Local + Cloud)

Combined annual and multi-year cost for complete SCADA system operation.

| Duration | Low | Medium | High |
|----------|-----|--------|------|
|----------|-----|--------|------|

|                |            |            |             |
|----------------|------------|------------|-------------|
| <b>1 Year</b>  | ₹ 66,000   | ₹ 1,30,300 | ₹ 2,15,400  |
| <b>2 Years</b> | ₹ 1,32,000 | ₹ 2,60,600 | ₹ 4,30,800  |
| <b>3 Years</b> | ₹ 1,98,000 | ₹ 3,90,900 | ₹ 6,46,200  |
| <b>4 Years</b> | ₹ 2,64,000 | ₹ 5,21,200 | ₹ 8,61,600  |
| <b>5 Years</b> | ₹ 3,30,000 | ₹ 6,51,500 | ₹ 10,77,000 |

## 5. Detailed Year-wise Breakdown

Cumulative costs split between Local and Cloud for each tier over 5 years.

| Year          | Low (Local) | Low (Cloud) | Med (Local) | Med (Cloud) | High (Total) |
|---------------|-------------|-------------|-------------|-------------|--------------|
| <b>Year 1</b> | ₹ 53,000    | ₹ 13,000    | ₹ 1,03,000  | ₹ 27,300    | ₹ 2,15,400   |
| <b>Year 2</b> | ₹ 1,06,000  | ₹ 26,000    | ₹ 2,06,000  | ₹ 54,600    | ₹ 4,30,800   |
| <b>Year 3</b> | ₹ 1,59,000  | ₹ 39,000    | ₹ 3,09,000  | ₹ 81,900    | ₹ 6,46,200   |
| <b>Year 4</b> | ₹ 2,12,000  | ₹ 52,000    | ₹ 4,12,000  | ₹ 1,09,200  | ₹ 8,61,600   |
| <b>Year 5</b> | ₹ 2,65,000  | ₹ 65,000    | ₹ 5,15,000  | ₹ 1,36,500  | ₹ 10,77,000  |

## 6. Important Notes

- Low tier assumes minimal manpower, basic internet, and limited spare inventory.
- Medium tier includes dedicated operator, better connectivity, and proactive maintenance.
- High tier covers 24/7 support, premium cloud hosting, comprehensive AMC with SLA.
- All costs are in Indian Rupees (INR) and are approximate estimates.
- GST and other statutory charges are not included and will be applicable as per government norms.
- Cloud hosting costs may vary based on actual usage, data volume, and provider pricing changes.
- AMC includes periodic calibration of sensors, PLC health check, and SCADA software updates.

## 7. What's Included

- 24/7 Real-time monitoring of all instruments (Level, Pressure, Flow, Quality)
- Alarm & alert system via SMS/Email for abnormal conditions
- Historical data logging with 7-day trends and analytics
- Pump runtime analytics and health monitoring
- Cloud-based dashboard accessible from anywhere
- Periodic maintenance and calibration of field instruments
- Software updates and security patches

## 8. Contact

For any queries regarding this cost estimation, please contact:

Orbit Automation

Email: [info@orbitautomation.com](mailto:info@orbitautomation.com)

Phone: Contact your project coordinator